

From lateral parabolic Schiffer to classical Schiffer: a spectral reduction

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1 Statement

The previous note (`direction2-heat-extension-computation.tex`) showed that the lateral parabolic Schiffer condition on a weighted pair (Ω, V) is strictly stronger than stationary weighted Schiffer, and identified the second-order obstruction $\eta_2|_{\partial\Omega} = 2\text{tr}(\nabla^2 V \nabla^2 \phi) + (\nabla V)^\top \nabla^2 \phi (\nabla V)$. The natural next step was to compute the higher residues η_k and test what the joint vanishing imposes. The spectral approach below sums the entire tower at once and produces the following clean reduction.

Theorem 1. *Let $\Omega \subset \mathbb{R}^n$ be a smooth bounded domain, $V \in C^\infty(\overline{\Omega})$, and $(\Omega, V, \phi, \lambda)$ a smooth weighted Schiffer pair:*

$$L_{\Omega, V} \phi = \lambda \phi \text{ in } \Omega, \quad \partial_\nu \phi = 0 \text{ on } \partial\Omega, \quad \phi|_{\partial\Omega} = c, \quad \phi \not\equiv 0.$$

Let $h = h_{\Omega, V}$ be the heat extension on $\Omega \times [0, 1]$ defined by (2.11) of arXiv:2412.06344 (unweighted Laplacian, Neumann lateral data, initial data ϕ). If

$$h(x, t) \equiv c \quad \text{for all } (x, t) \in \partial\Omega \times [0, 1],$$

then λ is a Neumann eigenvalue of the unweighted Laplacian $-\Delta$ on Ω , and the unweighted Neumann spectral projection $u := P_\lambda \phi$ is a nontrivial classical Schiffer overdetermined eigenfunction:

$$-\Delta u = \lambda u \text{ in } \Omega, \quad \partial_\nu u = 0 \text{ on } \partial\Omega, \quad u|_{\partial\Omega} = c.$$

The conclusion is striking: lateral parabolic Schiffer for the *weighted* pair (Ω, V) implies classical Schiffer for the *unweighted* Laplacian on the *base* domain Ω alone, with the potential V playing no role in the limit object beyond having selected an eigenvalue λ that lies in both spectra.

2 Proof

Throughout, Ω is bounded with smooth boundary, V smooth on $\overline{\Omega}$, and we use the unweighted Neumann Laplacian Δ_N on Ω with $L^2(dx)$ -orthonormal eigenbasis $\{\psi_j\}_{j \geq 0}$ and eigenvalues $0 = \mu_0 < \mu_1 \leq \mu_2 \leq \dots \rightarrow \infty$, so $-\Delta \psi_j = \mu_j \psi_j$ and $\partial_\nu \psi_j = 0$ on $\partial\Omega$.

Since Ω is bounded and $V \in C^\infty(\overline{\Omega})$, the measures dx and $e^{-V} dx$ are equivalent: $L^2(dx) = L^2(e^{-V} dx)$ as sets with equivalent norms. The eigenfunction ϕ admits a unique $L^2(dx)$ expansion

$$\phi = \sum_{j \geq 0} \alpha_j \psi_j, \quad \alpha_j = \int_{\Omega} \phi \psi_j dx.$$

Step 1: explicit heat extension. The bounded solution of

$$\partial_t h = \frac{1}{8}(\Delta_x h + \lambda h), \quad \partial_\nu h = 0, \quad h(x, 0) = \phi(x)$$

on $\Omega \times (0, 1)$ is, by separation of variables in the Neumann basis,

$$h(x, t) = \sum_{j \geq 0} \alpha_j e^{t(\lambda - \mu_j)/8} \psi_j(x). \quad (1)$$

The series converges in $C^\infty(\bar{\Omega} \times [0, 1])$ because $\phi \in C^\infty(\bar{\Omega})$ (so α_j decay faster than any power), at most finitely many μ_j are smaller than λ , and the rest produce exponentially decaying factors.

Step 2: collapse on the lateral boundary. Restrict (1) to $x \in \partial\Omega$ and group terms by distinct Neumann eigenvalue. Let $\Sigma := \{\mu_j : j \geq 0\}$ be the set of distinct Neumann eigenvalues of $-\Delta$ on Ω , and for each $\mu \in \Sigma$ let

$$f_\mu(x) := \sum_{j: \mu_j = \mu} \alpha_j \psi_j(x)|_{\partial\Omega} = (P_\mu \phi)(x)|_{\partial\Omega},$$

where P_μ is the orthogonal projection onto the μ -eigenspace of Δ_N . Then on $\partial\Omega \times [0, 1]$,

$$h(x, t)|_{\partial\Omega} = \sum_{\mu \in \Sigma} e^{t(\lambda - \mu)/8} f_\mu(x). \quad (2)$$

Assume the lateral parabolic Schiffer hypothesis $h|_{\partial\Omega \times [0, 1]} \equiv c$. Then for every $x \in \partial\Omega$,

$$\sum_{\mu \in \Sigma} e^{t(\lambda - \mu)/8} f_\mu(x) = c \quad \text{for all } t \in [0, 1]. \quad (3)$$

Step 3: linear independence of the exponential profiles. The family $\{t \mapsto e^{t(\lambda - \mu)/8} : \mu \in \Sigma\}$ is linearly independent in $C([0, 1])$: distinct exponentials are linearly independent (e.g. by repeated differentiation and the non-vanishing Vandermonde). For each $x \in \partial\Omega$, (3) thus forces

$$f_\mu(x) = \begin{cases} c & \mu = \lambda, \\ 0 & \mu \neq \lambda. \end{cases} \quad (4)$$

Two sub-cases.

Case A: $\lambda \notin \Sigma$. Then (4) reads $f_\mu \equiv 0$ on $\partial\Omega$ for every $\mu \in \Sigma$ and $c = 0$. Hence $\phi|_{\partial\Omega} = \sum_\mu f_\mu \equiv 0$, and combined with $\partial_\nu \phi = 0$ on $\partial\Omega$ this gives zero Cauchy data for the second-order uniformly elliptic equation $L_{\Omega, V} \phi = \lambda \phi$. By Holmgren / Carleman unique continuation across a smooth boundary, $\phi \equiv 0$ in Ω , contradicting non-triviality.

Case B: $\lambda \in \Sigma$. Then λ is a Neumann eigenvalue of $-\Delta$ on Ω , $f_\lambda \equiv c$ on $\partial\Omega$, and $f_\mu \equiv 0$ on $\partial\Omega$ for $\mu \neq \lambda$.

Step 4: extracting the classical Schiffer eigenfunction. In Case B, set $u := P_\lambda \phi$. By construction, u is a finite linear combination of Neumann eigenfunctions of $-\Delta$ at eigenvalue λ , hence

$$-\Delta u = \lambda u \quad \text{in } \Omega, \quad \partial_\nu u = 0 \quad \text{on } \partial\Omega.$$

Its boundary trace is $u|_{\partial\Omega} = f_\lambda \equiv c$. Since $c \neq 0$ (Case A is excluded), $u \not\equiv 0$. Thus u is a nontrivial classical Schiffer overdetermined eigenfunction on Ω , proving the theorem. $\square$

3 Corollaries

Corollary 2 (Radial test on the disc). *Let $\Omega = B \subset \mathbb{R}^n$ be the ball, $V \in C^\infty([0, R])$ a radial convex potential. Then no smooth nontrivial pair (B, V, ϕ, λ) with V nonconstant satisfies lateral parabolic Schiffer.*

Proof. For radial V , the Langevin operator preserves the radial sector and the radial ϕ exists with $-\phi'' - (n-1)r^{-1}\phi' + V'(r)\phi' = \lambda\phi$ on $(0, R)$, $\phi'(R) = 0$. The Schiffer overdetermination $\phi(R) = c$ is automatic by radiality. By Theorem 1, if lateral parabolic Schiffer holds, then $u = P_\lambda\phi$ is a classical Schiffer eigenfunction of $-\Delta$ at λ on B . The classical Schiffer overdetermined eigenfunctions on the ball are exactly the *radial* Neumann eigenfunctions, with eigenvalue λ a radial Neumann eigenvalue of $-\Delta$ on B . So λ is a radial Neumann eigenvalue of $-\Delta$, AND λ is the first Langevin eigenvalue of $L_{B,V}$. Subtracting the two radial ODEs satisfied by ϕ (as a Langevin eigenfunction) and by u (as a $-\Delta$ eigenfunction): with both having the same $\phi = u + \phi^\perp$ structure, the off-spectrum component $\phi^\perp$ satisfies zero Cauchy data on ∂B , hence vanishes by unique continuation (it solves $(-\Delta + V'\partial_r - \lambda)\phi^\perp = -V'u'$ with zero Cauchy data). Thus $\phi = u$. But then $L_{B,V}u = -\Delta u + V'(r)u' = \lambda u + V'(r)u'$, which equals λu only if $V'(r)u'(r) \equiv 0$. Since u is a nontrivial real-analytic radial eigenfunction with $u'(R) = 0$ but $u' \not\equiv 0$, the zero set of u' is discrete; hence $V' \equiv 0$, i.e., V is constant. Contradiction. $\square$

Corollary 3 (Generic non-coincidence). *If the Langevin first eigenvalue $\lambda_{\Omega,V}$ is not a Neumann eigenvalue of the unweighted Laplacian $-\Delta$ on Ω , then no smooth nontrivial ϕ produces a lateral parabolic Schiffer triple $(\Omega, V, \phi, \lambda_{\Omega,V})$. In particular, for any fixed convex Ω and any one-parameter family $\{V_s\}$ with $V_0 \equiv 0$ and $\partial_s V|_{s=0} \not\equiv 0$, the set of s for which lateral parabolic Schiffer can hold is discrete in s .*

Proof. The first statement is Case A of the theorem. The second follows because $s \mapsto \lambda_{\Omega,V_s}$ is real-analytic with nonzero first-order shape derivative under generic perturbations, while the Neumann spectrum of $-\Delta$ on Ω is fixed and discrete; so λ_{Ω,V_s} matches an unweighted Neumann eigenvalue only on a discrete set of s . $\square$

4 Reading the theorem from three angles

Reduction. Lateral parabolic Schiffer for weighted pairs on Ω reduces to (and is in fact equivalent up to the choice of V matching the eigenvalue, given Case B) classical Schiffer for the unweighted Laplacian on Ω at some eigenvalue. The potential V adds no new geometric content; it can only resonate with an existing Neumann eigenvalue of $-\Delta$.

Barrel dimension-reduction. Combined with the cap obstruction of the previous note (no barrel-lifted high-dim convex Schiffer counterexample), the new theorem gives a sharper statement. If a sequence of high-dimensional convex *approximate* Schiffer counterexamples Ω_d has the barrel form $\Omega_d = F_d(\Omega, V)$, then in the limit $d \rightarrow \infty$ they descend to a classical Schiffer overdetermined eigenfunction on Ω . In particular, *any hypothetical high-dim convex Schiffer counterexample of barrel form produces a classical Schiffer counterexample on the base Ω* . There is no high-codimensional escape via the barrel route.

What about Direction 2 itself? Direction 2 was: limits of convex Schiffer counterexamples. Theorem 1 narrows the admissible limits among the enlarged class of weighted pairs (Ω, V) . Specifically, the only way a sequence of weighted Schiffer counterexamples can have a heat-extension

parabolic-Schiffer behaviour is if the underlying base Ω already supports classical Schiffer overdetermined data. So if classical Schiffer is known on a class of base domains (low eigenvalues, centrally symmetric, etc.), the corresponding weighted/parabolic Schiffer classes are automatically empty over those bases.

5 What this leaves open

1. **The cap was the wrong target.** The reduction shows that the lateral parabolic Schiffer condition is automatically very rigid; the cap obstruction (Prop. 6 of the previous note) was the easy part. The real geometric content of weighted Schiffer is *stationary* weighted Schiffer, which sits outside the lateral parabolic class. Concretely: a stationary weighted Schiffer pair (Ω, V) at a Langevin eigenvalue $\lambda_{\Omega, V}$ not in the Neumann spectrum of $-\Delta$ exists without contradicting Theorem 1, but the heat extension of its eigenfunction simply does not behave well on the lateral boundary.
2. **Stationary weighted Schiffer is still the right target for Strategy A.** The augmented compactness program of Strategy A of the original note now reads: enumerate stationary weighted Schiffer pairs (Ω, V) and use *stationary* overdetermined rigidity, not parabolic Serrin, to rule them out. Parabolic Serrin only applies to the (much smaller) subclass where $\lambda_{\Omega, V}$ coincides with an unweighted Neumann eigenvalue.
3. **The barrel construction has the wrong target.** The barrel limit produces an unweighted parabolic problem (1). To capture stationary weighted Schiffer the right limiting object would be the *weighted* elliptic problem on Ω , which is what one would obtain via a warped barrel as suggested at the end of the previous note. Implementing this should be the next concrete construction.
4. **A clean open question.** Does there exist a smooth convex pair (Ω, V) with V non-constant for which a stationary weighted Schiffer eigenfunction ϕ exists? Equivalently: does the weighted Pompeiu property hold for all (Ω, V) with nonconstant V ? This is the natural “log-concave Schiffer conjecture” aligned with the de Dios Pont log-concave extension of the hot spots conjecture.